

3) $|x_2 + x_4| \leq |x_2| + |x_4|$ (By Triangle inequality)
 $\therefore$ The condition 1 of the defn. of linear transformation is violated since
 $T(x_1 + x_3, x_2 + x_4) \neq T(x_1, x_2) + T(x_3, x_4)$ $\square$

33. Show that the transformation T defined by $T(x_1, x_2) = (2x_1 - 3x_2, x_1 + 4, 5x_2)$ is not linear.

Solution: $T(cx_1, cx_2) = (2cx_1 - 3cx_2, cx_1 + 4, 5cx_2)$
 $\neq c \cdot T(x_1, x_2) = c \cdot (2x_1 - 3x_2, x_1 + 4, 5x_2) = (2cx_1 - 3cx_2, cx_1 + 4c, 5cx_2)$ (Scalar multiplication)

$\therefore T(cx_1, cx_2) \neq c \cdot T(x_1, x_2)$ for all $c \in \mathbb{R}$

The condition 2 of the defn. of linear transformation is violated. $\square$

34. Let $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a linear transformation. Show that if T maps two linearly independent vectors onto a linearly dependent set, then the equation $T(x) = 0$ has a nontrivial solution.

Solution: Let p and q be two linearly independent vectors. Also suppose $T(p)$ and $T(q)$ are linearly dependent. $\therefore$ By defn. of linear dependence,

$c_1 p + c_2 q = 0$ if and only if $c_1 = c_2 = 0$.
 By defn. of linear dependence, there exists a solution where both c_3 and c_4 are not zero together s.t. $c_3 T(p) + c_4 T(q) = 0$.

$\Rightarrow T(c_3 p) + T(c_4 q) = 0$ (Condition 2) $\Rightarrow T(c_3 p + c_4 q) = 0$ (Condition 1)

Let $x = c_3 p + c_4 q$. $\therefore T(x) = 0$. $\therefore$ Both c_3 and c_4 are not zero together, a nontrivial solution exists. $\square$

35. Let $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be the transformation that reflects each vector $x = (x_1, x_2, x_3)$ through the plane $x_3 = 0$ onto $T(x) = (x_1, x_2, -x_3)$. Show that T is a linear transformation.

Solution: $T(x) = (x_1, x_2, -x_3)$ $T(y) = (y_1, y_2, -y_3)$
 $T(x+y) = (x_1+y_1, x_2+y_2, -(x_3+y_3)) = (x_1, x_2, -x_3) + (y_1, y_2, -y_3)$ (Vector addition)
 $= T(x) + T(y)$. $\therefore$ Condition 1 satisfied